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(Bottom's up)

(forward)



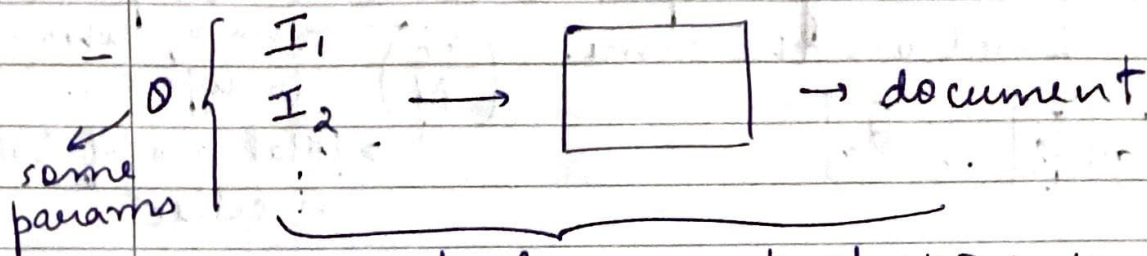
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- no forward
models
req.

→ more nuances, more power

Generative	Discriminative
- learns to gen. data - $P(X, Y) \rightarrow$ joint - models the distribuⁿ of each class - gen. new data	- learns to sep. classes - $P(Y X) \rightarrow$ condi. - draws decision boundary btw. each class - usually doesn't gen. new data.

- there are many design philosophies in spell check, bayesian model is one of them.



more or less most of NLP tasks work in this way only. It is impt. to observe the data pts. & capture the underlying patterns. critical ML tasks $\rightarrow$ inference & predicⁿ

modeling problem
 what is θ ?
 distribuⁿ of θ

given θ , can
 I predict sth.
 based on that θ .

- detecⁿ problem - I have a word, not encountered in dictionary (context-insensitive)
- correctⁿ problem -
 - ① identify the candidates
 - ② value of prior
 - ③ # likelihoods & posterior
- how to identify the candidates? $\rightarrow$ bigrams (efficient coefficients) + inverted index

apple $\rightarrow$ -a, ap, pp, pl, le, e-

aple $\rightarrow$ -a, ap, pl, le, e-

p is missing

this helps to calculate

we will get exhaustive list of candidates

- priors - eg. $\text{teh} \mid \text{the} \Rightarrow 99\%$ of time

take a big corpus, & calculate the priors based on the same.

$d_1 \quad d_2 \quad \dots \quad d_n$

$$\left(\frac{n_c}{N} \right)$$

no. of tokens to type c .

total no. of tokens in collecⁿ

no. of types is no. of distinct words you see

What could be potential problem of priors, calculating in this way?

- dependent on the corpus only.
- there is some word in dictionary, but not in corpus, that means prior is 0.
↳ smoothing